Mohammad Ninad Mahmud Nobo
Bangladesh University of Engineering and Technology (BUET) | CSE

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Summary

Computer Science graduate specialized in Artificial Intelligence, Machine Learning, Large Language Models, and software engineering across research and industry. Skilled in building production-grade software systems, including deep learning models, multi-agent AI pipelines, and web applications. Research spans LLM-based software testing, medical AI and applied machine learning.

Education

Bangladesh University of Engineering and Technology (BUET) **2022 – 2026**
Bachelor of Science in Computer Science and Engineering (CSE) **CGPA: 3.60 / 4.00**
Relevant Coursework: Data Structures and Algorithms, Algorithm Engineering, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Compiler Design, Artificial Intelligence, Machine Learning, Computer Graphics
Research Interests: Large Language Model Systems, AI-Assisted Software Engineering, Applied Machine Learning, Medical AI

Professional Experience

- **AI Software Engineer**
  Department of Research and Development (R&D) - Saturn Textiles Limited *2026 – Present*
  Web Development [Saturn R&D Website | GitHub] [FABINS Automation Website | GitHub]
  Next.js • React • Spring Boot • REST APIs • Brevo • GitHub Actions *2026*
  – Developed and deployed Saturn R&D Website and FABINS Product Portfolio using Next.js/React with responsive interfaces.
  – Built Spring Boot REST APIs for form processing and email notifications, with GitHub Actions for CI/CD and deployment.
  FABINS - Fabric Inspection Automation | ML & Computer Vision
  Python • PyTorch • Computer Vision • Deep Learning • Image Processing • Industrial Automation *2026 – Present*
  – Developing computer vision and deep learning pipelines for fabric defect detection, classification, and quality inspection.

Research Experience

- **AutoTestGenX: Multi-Agent LLM Framework for End-to-End Web Testing (Undergraduate Thesis)** [GitHub]
  Python • LLM Engineering • Multi-Agent Systems • Web Testing • Browser Automation *2025 – 2026*
  – Designed a multi-agent LLM framework to automate web test generation and execution from natural-language requirements.
  – Built benchmark datasets and evaluated against ground-truth suites, achieving 84.0% scenario coverage and 90% error detection.
- **MedCAR: Conflict-Aware Medical Reasoning System** [GitHub]
  Python • Deep Learning • Medical AI *2026*
  – Built a conflict-aware multi-agent system for chest X-ray reasoning that detects and resolves contradictory model outputs.
  – Achieved 79.0% conflict detection accuracy on 503-case ChestAgentBench-X benchmark using trust calibration and abstention.
- **Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization** [arXiv : 2602.14291]
  Python • Speech Processing • Automatic Speech Recognition • Speaker Diarization *2026*
  – Annotated Bangla audio data for Automatic Speech Recognition (ASR) and speaker diarization for Bengali-Loop benchmark.

Academic Projects

- **MindTrace: AI-Powered Dementia Care Application** [GitHub | Feature | Infrastructure]
  Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure *2025*
  – Developed an application for dementia patients and caregivers with personalized assistance and caregiver support features.
  – Designed and implemented the Android UI in Kotlin with Spring AI integration for intuitive and personalized AI interactions.
- **Gemma VetCare: AI-Assisted Farming Support System** [GitHub | Feature]
  Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB *2025*
  – Built an application for real-time veterinary diagnosis and treatment, with RESTful APIs for low-connectivity environments.
- **TCP Window Scaling Attack** [GitHub]
  Python • TCP/IP • Network Security • Wireshark • Linux *2025*
  – Implemented a three-VM setup for ARP poisoning and TCP window scaling attacks, with defense and Wireshark analysis.
- **Computer Graphics Pipeline** [GitHub]
  C++17 • OpenGL/GLUT • Rasterization • Ray Tracing *2025*
  – Implemented graphics pipeline with vector/matrix transformations, OpenGL, rasterization, Z-buffering, ray tracing, and PBR.
- **End-to-End Compiler Construction** [GitHub]
  C++17 • Flex • Bison/Yacc • 8086 Assembly *2024*
  – Built a pipeline with symbol tables, lexical, syntax, and semantic analysis, and 8086 code generation using Flex and Bison/Yacc.
- **Smart Gardening Automation** [GitHub | Demo]
  Arduino • Embedded C++ • ATmega32 • DHT11 • Soil/Rain Sensors • Bluetooth • Stepper Motor *2024*
  – Developed a gardening system with sensing, watering, fertilizer/pesticide control, smart shade control, and remote monitoring.

Technical Skills

AI/ML: PyTorch, TensorFlow, Scikit-learn, Large Language Models, Prompt Engineering, Computer Vision, Image Processing
Backend & Full-Stack: Spring Boot, Spring AI, Next.js, React, Node.js, Express.js, REST APIs, Android, Multi-Agent Systems
Testing & Automation: Software Testing, Web Testing, Browser Automation, Industrial Automation
DevOps & Tools: Git, Docker, Docker Compose, GitHub Actions, Firebase, Azure
Languages: Python, C/C++, Java, JavaScript, Kotlin, HTML, CSS, SQL
Systems & Networking: Linux, TCP/IP, Wireshark, 8086 Assembly
Databases: PostgreSQL, MongoDB, Redis